
Before Health Agents Interpret Movement: Lessons from a Gait Representation Study

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Abstract

1 We argue that movement summaries supplied to future health assistants should
2 preserve what was measured, how recordings were weighted, and which uses
3 were tested. A JEPA-style gait case illustrates why. For the same 64 normal-
4 annotated validation clips, feature similarity across training stages averages to
5 0.89 with equal clip weights and 0.70 with equal video weights; one video sup-
6 plies 60 clips. Neither value establishes retained predictive ability or a change
7 in patient health. An exploratory comparison also favors simple pose summaries
8 over learned features on 20 held-out videos. These observations motivate an evi-
9 dence record accompanying movement summaries and a proposed test of its value
10 for interpretation. No generative assistant or clinical intervention was evaluated.

11 1 Position and clinical relevance

12 A health assistant could eventually combine video-derived movement summaries with other patient
13 information to support mobility review. A similarity score can compare a fixed recording across two
14 model versions, rather than a person’s movement across two visits. Repeated clips can also make
15 one recording dominate an apparently broad evaluation.

16 Our position is that the measurement and evaluation conditions should remain attached to movement
17 results when they are passed downstream. Model Cards already address intended uses and evaluation
18 limits [5], and the V3 framework distinguishes stages of validation for digital measurements [4]. We
19 apply those principles to a concrete gait representation case, concentrating on recording weights and
20 the distinction between model coordinates and useful function. The encoder uses a joint-embedding
21 predictive architecture inspired by S-JEPA [1]; it does not implement a clinical agent.

22 2 Case study and observations

23 The study uses 639 quality-screened clips from 97 YouTube videos drawn from GAVD, a manually
24 curated resource for clinical gait analysis [6]. Its selected annotations cover normal gait and Parkin-
25 son’s, stroke, myopathic, and cerebral-palsy categories, without independent diagnostic verification
26 here. All clips from a video share its role: 59 videos for training, 18 for validation, and 20 for
27 testing. The evidence comes from one split and one training initialization. Video separation does
28 not guarantee person separation across uploads.

29 A compact encoder receives 33-landmark poses resized to 64 frames and predicts hidden features
30 at selected body landmarks. Training begins with normal-annotated clips, then cumulatively adds
31 the other categories. Category labels guide that order but are not targets in the encoder loss. We
32 reanalyze retained predictions and similarity summaries, without rerunning encoder training; some
33 runtime settings were not fully recorded.

34 **Holding similarities fixed, changing the weights.** For each normal-annotated validation clip, co-
 35 sine compares its features after normal-only training with its features after final training. There are
 36 64 clips from five videos, with one video contributing 60 clips. Equal clip weighting assigns that
 37 video about 94% of the total weight and gives mean cosine 0.89. Averaging first within videos and
 38 then equally across videos assigns it 20% and gives 0.70. The same recorded similarities enter both
 39 calculations.

40 These are different summaries of the recordings, not competing estimates of patient stability. Equal
 41 video weights are not universally preferable and do not imply equal patient weights. Moreover, either
 42 average can change as model coordinates change without establishing whether useful movement
 43 information was lost. Functional retention needs an actual prediction task assessed across model
 44 versions.

45 **Testing a simpler input baseline.** Three separately fitted logistic-regression classifiers predict the
 46 dataset annotations using pose summaries, frozen learned features, or landmark availability without
 47 coordinates. Validation selects regularization, after which classifiers are refitted on the 77 training
 48 and validation videos. The same 20 test videos give:

Table 1: Descriptive results from one split and initialization. Balanced accuracy averages within-category recall over the five annotations.

Input	Correct videos	Balanced accuracy
Pose and movement summaries	10/20	0.44
Learned pose features	6/20	0.26
Landmark availability only	6/20	0.25

49 All three classifiers misclassify the three stroke-annotated test videos. Two categories have only
 50 two test videos each. Selection also uses averaged features per video, whereas testing averages the
 51 category probabilities predicted for its clips; these procedures need not agree. Without repeated runs
 52 and a matched untrained encoder, the result supports retaining a pose baseline but cannot isolate the
 53 effect of pretraining or establish a general ranking.

54 3 Proposal, counterarguments, and next evaluation

55 We propose that each movement summary include its measured quantity, reference model, record-
 56 ing unit and weights, observed result, and interpretation limits. For the weighting example, that
 57 record would state: fixed clips compared across training stages; five equally weighted videos; mean
 58 cosine 0.70; predictive retention and patient mobility change untested. This specializes established
 59 reporting guidance to the result being interpreted, rather than proposing a new general validation
 60 framework.

61 A follow-up could compare the same assistant’s interpretations with a bare score against interpre-
 62 tations with this record, using blinded review of unsupported mobility claims and recognition of
 63 insufficient evidence. Tests should distinguish changes in recordings from model updates and inde-
 64 pendently measured movement changes. The record’s benefit remains a hypothesis; documentation
 65 may be ignored or add burden. Clinical usefulness requires a defined workflow and evaluation of
 66 safety and human factors [7].

67 A larger model or better evaluation could favor learned representations without eliminating the need
 68 to state what a summary measures. Forecasting remains untested and would require inputs and
 69 preprocessing restricted to the observed past.

70 The public-video sample, uncertain cross-upload identities, estimated pose geometry, and incom-
 71 plete training records limit generalization. GAVD’s annotation repository uses the MIT License;
 72 separately hosted videos have additional access and use conditions [3, 2]. Project-specific ethics
 73 and data-use reviews remain unresolved, and no approval or exemption is claimed. The evidence
 74 supports a focused position about interpreting movement inputs, with agent behavior and patient
 75 benefit left for subsequent evaluation.

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